



FACULTY OF MECHANICAL TECHNOLOGY AND ENGINEERING

Project Report Cover Sheet
for Submission of Group Work

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ABSTRACT

This project entails the design and development of a prototype scooter lawn mower aimed at improving performance, efficiency, and user convenience. Through systematic engineering processes, the prototype integrates sustainability features, ergonomic design, and enhanced mobility for varied terrains. Key phases include conceptual design, material selection, CAD modelling, prototyping, and testing. Advanced simulations and empirical tests validated the structural integrity, cutting efficiency, and user-friendliness of the design. Challenges such as motor performance, cutting blade optimization, and traction on grass surfaces were addressed through innovative engineering solutions. The resulting prototype demonstrates significant potential for practical application and highlights areas for further refinement, including braking systems and energy recovery mechanisms.

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1.0 INTRODUCTION

1.1 Background of Study

The integrated design project provides opportunities for engineering students to apply theoretical knowledge and technical skills to the design and development of a practical, functional, and innovative product. The topic selected to be the title of the integrated design project for year 2024 is lawn mower. Lawn mowers are essential tools in landscaping and maintenance, widely used in residential, commercial, and industrial applications. This project focuses on reimagining and optimizing traditional lawn mower designs to improve performance, efficiency, sustainability, and user convenience. By working on this project, students will gain valuable knowledge about mechanical design, electrical systems, material selection, and manufacturing processes.

In order to maintain a well functioned lawn mower, it requires a reliable and efficient cutting mechanism, and the lawn mower has evolved over the years to meet this demand. However, as society started to move toward sustainability and technological advancements, there is a growing need to develop eco-friendly and smart solutions for everyday tasks, for example, lawn care. This project encourages students to think critically about the challenges faced in lawn maintenance and to explore modern technologies, such as automation, renewable energy sources, and smart sensors to create innovative solutions.

The objective of this project is to design and construct a lawn mower prototype that balances in functionality, cost-effectiveness, and environmental impact. In this project, students are encouraged to explore various power sources, such as electric motors, battery systems, or even solar energy in order to reduce carbon emissions and promote sustainability. Additionally, the project emphasizes ergonomic and user-friendly design to ensure that the finished product is safe, easy to operate, and accessible to a wide range of users.

Throughout the project, students will engage in the complete engineering design process, starting from problem definition and requirement analysis to conceptualization, CAD modelling, prototyping, and testing. They will apply principles of engineering mechanics to optimize the cutting and moving mechanisms, evaluate material properties to ensure durability and lightweight construction, and integrate electrical systems for powering the device. The

project also offers a platform to experiment with automation features, such as autonomous navigation or remote control, providing a glimpse into the future of smart lawn care systems.

Moreover, the lawn mower project serves as a collaborative learning experience, requiring teamwork, project management, and effective communication between the group members. Students will work together to solve real-world engineering problems, balancing technical feasibility, cost constraints, and environmental considerations. This project also helps to develop critical soft skills, such as time management, resource allocation, and documentation, which prepare students for professional challenges in their future careers.

In conclusion, the lawn mower project represents a comprehensive engineering challenge that bridges the gap between classroom learning and real-world application. By combining creativity, innovation, and technical expertise, students can design a solution that not only meets the needs of modern users but also contributes to a more sustainable future. This hands-on experience will help them build a solid foundation for their engineering careers and inspire them to tackle complex problems in the field of mechanical and electrical systems.

1.2 Problem Statement

Lawn mowers are essential for maintaining lawns, but they face several challenges that limit their efficiency, usability, and reliability, which offer an excellent opportunity for engineering innovation. In this project, one of the key issues is the frequent need to charge or replace energy supplies, which disrupts operation and reduces convenience. Electric mowers, while more sustainable, are prone to challenges such as rapid battery drainage and motor failures, which compromise their reliability and lifespan. Additionally, motor overheating due to poor airflow or inadequate cooling mechanisms is a common problem which will lead to reduced efficiency and potential damage to the motor. Besides, the lawn mower also faces mobility issues including flat tires, worn-out bearings, and damaged wheels, making it difficult to move the lawn mowers on uneven or rugged terrain. Furthermore, many lawn mowers are heavy and difficult to carry, which limits portability and user convenience. These issues highlight the need for an improved lawn mower design that addresses power efficiency, durability, overheating prevention, and ease of use. This project provides a valuable opportunity to apply their knowledge and skills to develop an innovative and practical solution that improves the functionality and user experience of lawn mowers.

1.3 Objectives

The objectives for this project are stated as below:

1. Explain and apply an appropriate design method at the particular design phase in the course of developing a practical solution of an engineering design problem
2. Develop practical design solution through a systematic investigation of the engineering design problem
3. Communicate effectively in written, oral and visual means in a technical setting

1.4 Scope of Project

The scopes of this project are stated as below:

1. Design and manufacture a functionable and innovative lawn mower either manual or electric lawn mower which meet the demand of the market, environmentally friendly and cost effective.
2. Analyze and optimize the performance of a lawn mower which focuses on power efficiency, cutting efficiency and durability of the lawn mower by benchmarking and selection of suitable components.

2.0 LITERATURE REVIEW AND BENCHMARKS

2.1 Literature review

The lawn mower was invented in 1830 by Edwin Beard Budding of Stroud, Gloucestershire, England. Budding's mower was designed primarily to cut the grass on sports grounds and extensive gardens, as a superior alternative to the scythe, and was granted a British patent on August 31, 1830. Budding's first machine was 19 inches (480 mm) wide with a frame made of wrought iron. The mower was pushed from behind. Cast-iron gear wheels transmitted power from the rear roller to the cutting cylinder, allowing the rear roller to drive the knives on the cutting cylinder; the ratio was 16:1. Another roller placed between the cutting cylinder and the main or land roller could be raised or lowered to alter the height of the cut. The grass clippings were hurled forward into a tray-like box. It was soon realized, however, that an extra handle was needed in front to help pull the machine along. Overall, these machines were remarkably similar to modern mowers.

Two of the earliest Budding machines sold went to Regent's Park Zoological Gardens in London and the Oxford colleges. In an agreement between John Ferrabee and Edwin Budding dated May 18, 1830, Ferrabee paid the costs of enlarging the small blades, obtained letters of patent and acquired rights to manufacture, sell and license other manufacturers in the production of lawn mowers. Without a patent, Budding and Ferrabee were shrewd enough to allow other companies to build copies of their mower under licence, the most successful of these being Ransomes of Ipswich, which began making mowers as early as 1832. His machine was the catalyst for the preparation of modern-style sporting ovals, playing fields (pitches), grass courts, etc. This led to the codification of modern rules for many sports, including for football, lawn bowls, lawn tennis and others.



Figure 2.1: An early cylinder (reel) mower

It took ten more years and further innovations to create a machine that could be drawn by animals, and sixty years before a steam-powered lawn mower was built. In the 1850s, Thomas Green & Son of Leeds introduced a mower called the Silens Messor (meaning silent cutter), which used a chain drive to transmit power from the rear roller to the cutting cylinder. These machines were lighter and quieter than the gear-driven machines that preceded them, although they were slightly more expensive. The rise in popularity of lawn sports helped prompt the spread of the invention. Lawn mowers became a more efficient alternative to the scythe and domesticated grazing animals.

Manufacture of lawn mowers took off in the 1860s. By 1862, Ferrabee's company was making eight models in various roller sizes. He manufactured over 5000 machines until production ceased in 1863. The first grass boxes were flat trays but took their present shape in the 1860s. James Sumner of Lancashire patented the first steam-powered lawn mower in 1893. His machine burned petrol or paraffin (kerosene) as fuel. These were heavy machines that took several hours to warm up to operating pressure. After numerous advances, these machines were sold by the Stott Fertilizer and Insecticide Company of Manchester and Sumner. The company they both controlled was called the Leyland Steam Motor Company.

Around 1900, one of the best known English machines was the Ransomes' Automaton, available in chain- or gear-driven models. Numerous manufacturers entered the field with petrol (gasoline) engine-powered mowers after the start of the 20th century. In 1902, The first was produced by Ransomes. JP Engineering of Leicester, founded after World War I, produced a range of very popular chain-driven mowers. About this time, an operator could ride behind animals that pulled the large machines. These were the first riding mowers.

The first United States patent for a reel lawn mower was granted to Amariah Hills on January 12, 1868. In 1870, Elwood McGuire of Richmond, Indiana designed a human-pushed lawn mower, which was very lightweight and a commercial success. John Burr patented an improved rotary-blade lawn mower in 1899, with the wheel placement altered for better performance. Amariah Hills went on to found the Archimedean Lawn Mower Co. in 1871. In the United States, gasoline-powered lawn mowers were first manufactured in 1914 by Ideal Power Mower Co. of Lansing, Michigan, based on a patent by Ransom E. Olds. Ideal Power Mower also introduced the world's first self-propelled, riding lawn tractor in 1922, known as the "Triplex". The roller-drive lawn mower has changed very little since around 1930. Gang

mowers, those with multiple sets of blades to cut a wider swath, were built in the United States in 1919 by the Worthington Mower Company.



Figure 2.2: The first gasoline-powered lawn mower, 1902



Figure 2.3: A horse drawn lawn mower on an Australian golf course

Today, lawn mowers are used in various locations, including public parks, landscaping services and golf course maintenance. This lawn mower has evolved over time, incorporating advanced technologies and designs to improve the performance and efficiency.

2.2 Benchmarks

The riding lawn mower project aspires to efficiently cut grass on large lawns by allowing the user to sit on the machine while it operates, significantly reducing physical effort compared to a walk-behind mower and making mowing larger areas faster and more comfortable.

- **Safety and Compliance**

Prioritizing user and environmental safety, the Safety and Compliance benchmark guides the integration of fail-safe mechanisms and designs. The goal is to not only meet but exceed industry safety standards, ensuring the well-being of both users and the ecosystems.

- **Efficiency and Effectiveness**

The backbone of performance lies in the Efficiency and Effectiveness benchmark. This standard drives continuous improvement in technology, striving to set the bar for waste collection that minimizes results while minimizing energy consumption and overall environmental impact.

- **Materials and Construction**

At the core of the project lies the Materials and Construction benchmark, setting standards for the selection of resilient components. The goal is to ensure the longevity of the lawn mower while minimizing its ecological footprint.

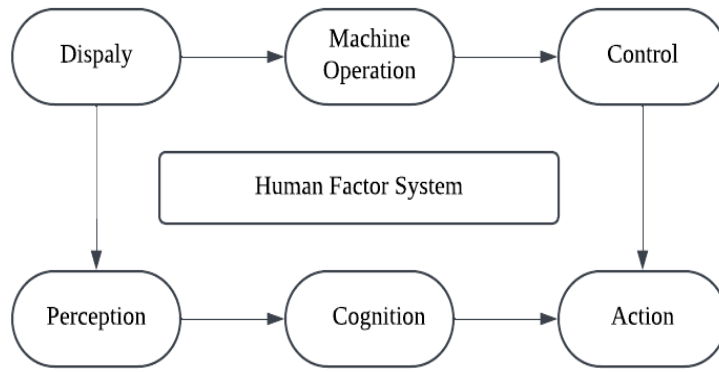
3.0 CONCEPT DESIGN DEVELOPMENT

The concept design phase focuses on overcoming the challenges identified in current lawn mower designs, as outlined in the problem statement. It aims to full-fill the project's objectives by prioritizing improved performance, enhanced sustainability and greater user convenience. Concept design development for improving a lawn mower project can be elaborated further as follows:

3.1 Human Factors Consideration

No	HUMAN FACTOR	DESCRIPTION
1.	Ergonomics	Design the system to optimize user comfort and reduce operator fatigue. Consider factors such as seating arrangements and control placement to ensure ergonomic interaction between the user and the system.
2.	User Interface Design	Develop an intuitive and user-friendly interface for operators to control the lawn mower. Use clear instructions and user centred design principles to enhance usability and minimize errors during operation.
3.	Safety Considerations	Incorporate safety features to protect operators and prevent accidents. Ensure compliance with safety standards and regulations.

Table 3.1: Human Factors Consideration



ASPECT	DESCRIPTION	IMPORTANCE
Perception	The ability of operator to perceive relevant information from the environment	To gather accurate and real time data about the lawn mower's surroundings, obstacles and operating conditions
Cognition	The mental processes involved in understanding, interpreting and making decisions based on information	To analyse the perceived information, interpret it correctly and make informed decisions during operation
Action	The physical movements and interactions made by the operator to control the lawn mower	To operate the lawn mower effectively and carry out the cutting tasks efficiently
Control	The mechanisms which the operator controls and uses the lawn mower	Intuitive and ergonomic control systems are essential for accurate and comfortable operation, ensuring the operator's control over the lawn mower's movement
Machine operation	The overall functioning and performance of the lawn mower as a machine	To ensure reliable and efficient operation, meeting the desired cutting objectives while minimizing breakdowns and maintenance needs
Display	The visual information presented to the operator	To provide vital information about the lawn mower's status

3.2 Application of DFMA Principles

DFMA stands for Design for Manufacturing and Assembly. It is a set of principles and guidelines that were used in product design to make manufacturing and assembly processes easier, faster, and more cost-effective. Table below shows the techniques that have been used to ensure that the lawn mower is efficient to make and assemble from the start.

TECHNIQUE	DESCRIPTION
Material Optimisation	● To select materials that are durable, cost-effective and suitable in producing the lawn mower. ● Use corrosion-resistant materials such as stainless steel or aluminum, for parts that are exposed to outdoor elements.
Simplification	● To minimise the number of components and reduce complexity in the design. ● Batteries for blade and scooter were combined and shared in one box to reduce the number of batteries used.
Standardization	● To use standardised parts and components that are easy to find and fit into the design. ● Choose motors, wheels and other parts that are already made and widely available in store.
Modular Design	● To divide the lawn mower into modular components that can be easily assembled and disassembled. ● Create separate modules for the motor, cutting blade system and handle assembly.

Table 3.2: Application of DFMA Principles

3.3 Implementation of FMEA

FMEA focuses on identifying potential points of failure in the design or operation of the riding lawn mower. This includes examining components and overall functionality to minimize risks such as breakdowns, accidents or inefficiencies. Table below shows an elaboration of FMEA of the riding lawn mower:

Failure Mode	Failure Effect	Cause	Occurrence	Detection	Severity	RPN	Corrective Action
Motor Failure	Inability to move or power on the lawn mower	Electrical component failure, mechanical issues, power supply problems	6	7	8	336	Implement regular maintenance, monitor motor performance, conduct through testing and quality checks during manufacturing

Failure Mode	Failure Effect	Cause	Occurrence	Detection	Severity	RPN	Corrective Action
Power Source Failure	Reduced performance or complete failure	Battery or power supply issues	3	6	9	162	Implement regular battery maintenance and monitoring, use reliable and high-quality power sources

Table 3.3: Application of FMEA

4.0 ENGINEERING ANALYSIS

4.1 Methods

The lawn mower project utilized surveys, simulations with ANSYS and SolidWorks, and case studies to guide the design process and ensure the product meet both user expectations and technical requirements. Surveys were conducted to gather valuable perceptions directly from potential users such as homeowners and landscaping professionals to understand their needs, preferences and frustrations with existing lawn mowers. The survey questions focused on identifying the average lawn size, preferred power source (gas, battery-powered, or manual), desired features such as mulching or grass collection, and common complaints such as noise, weight, or inefficiency. These surveys helped define the functional and performance requirements for the design.

Simulations played a crucial role in analyzing and optimizing the lawn mower design. **SolidWorks** was first used to create 3D CAD models of the lawn mower, including all major components such as the blade, frame, handle, and wheels. These models allowed the team to visualize the overall design and refine it iteratively before moving on to more detailed analyses. Once the design was finalized, **ANSYS** was used to perform advanced engineering simulations.

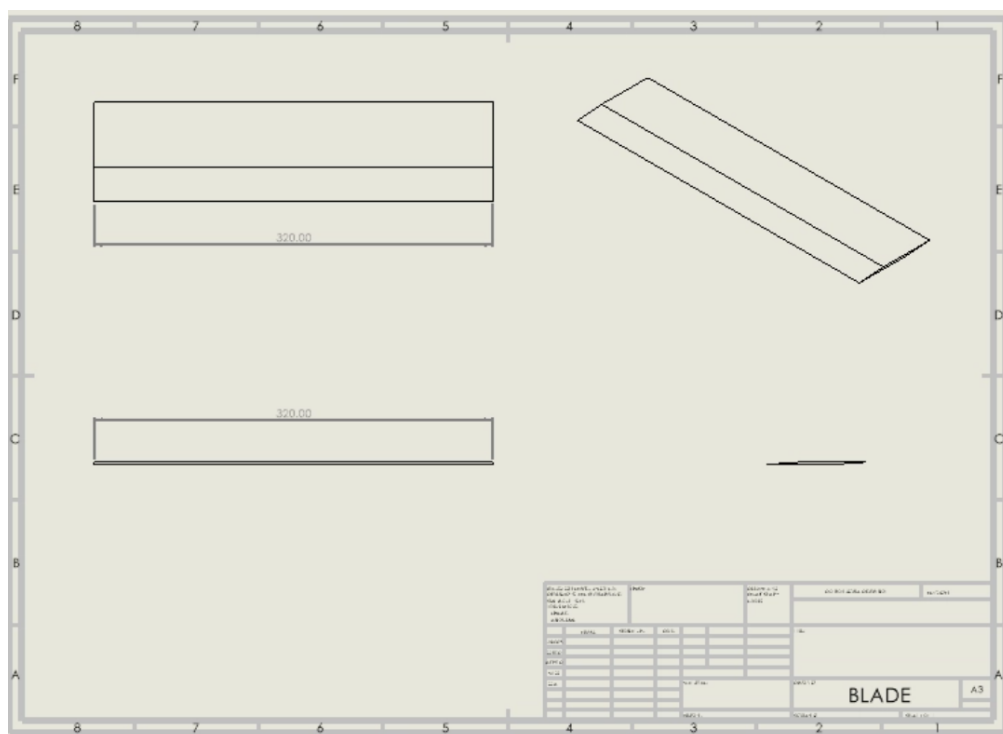


Figure 4.1: Drawing of blade in solidworks

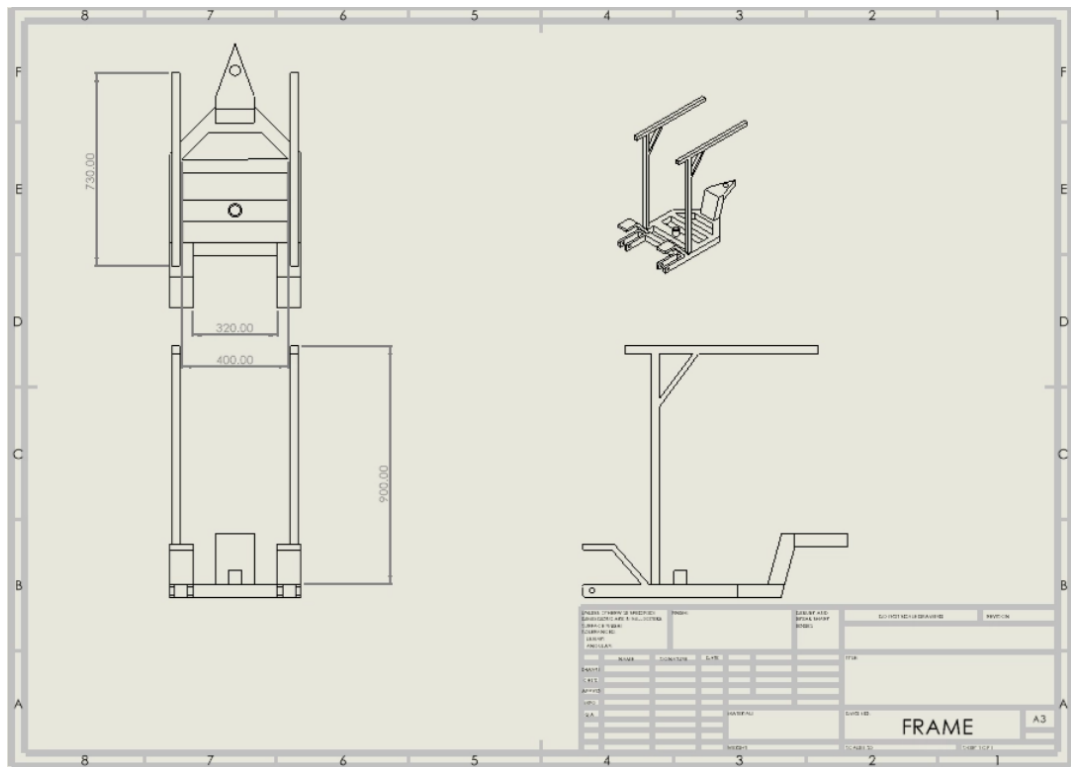


Figure 4.2: Drawing of frame in solidworks

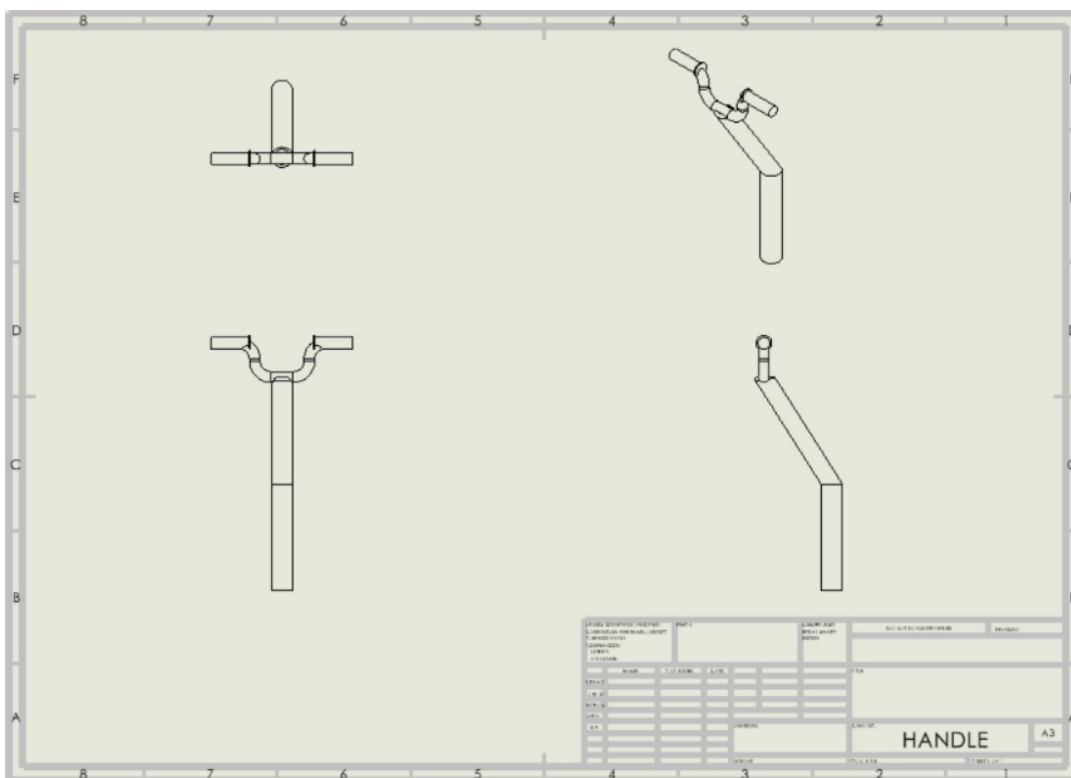


Figure 4.3: Drawing of handle in solidworks

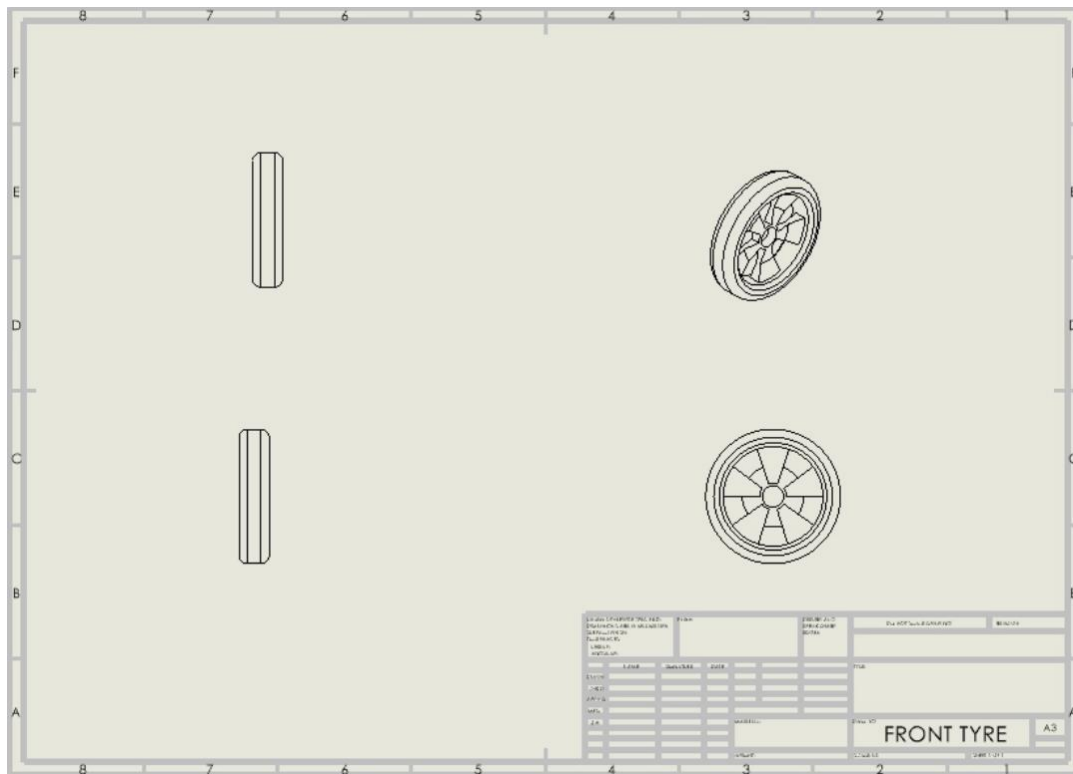


Figure 4.6: Drawing of front tyre in solidworks

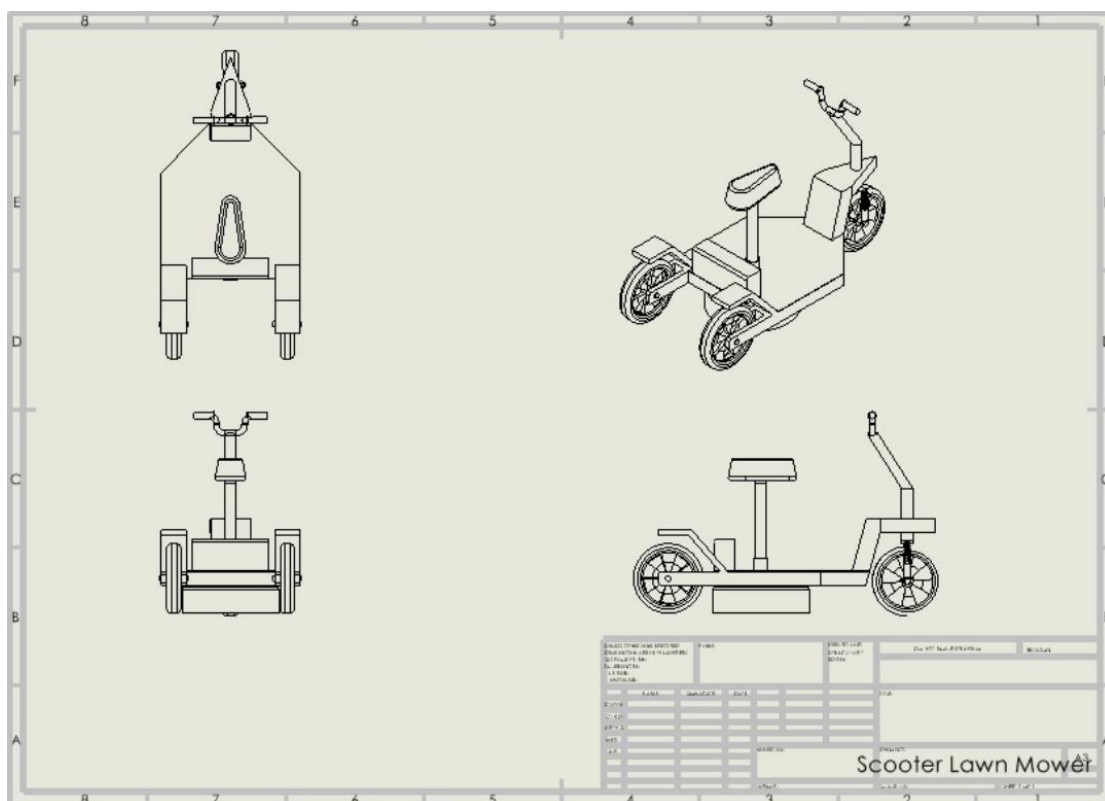
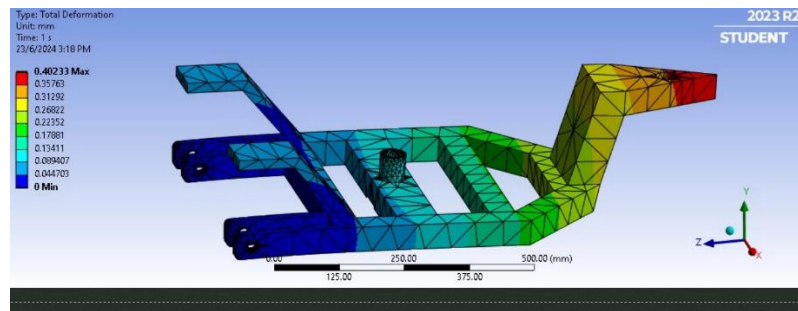


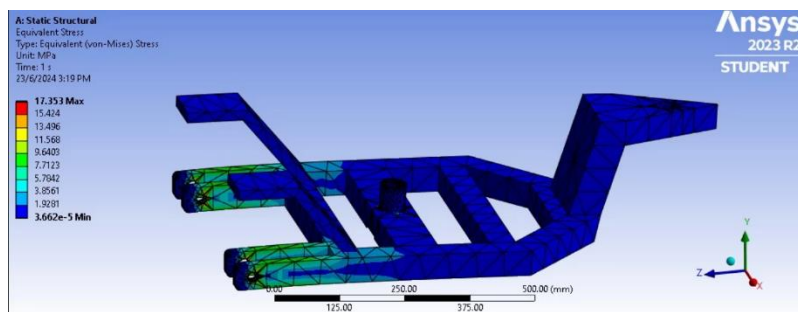
Figure 4.7: Drawing of riding lawn mower in solidworks

Finite Element Analysis (FEA) was conducted to evaluate the structural strength and durability of critical components under various loading conditions. For example, the were analyzed to ensure they could withstand the weight of the mower and operator forces during use.



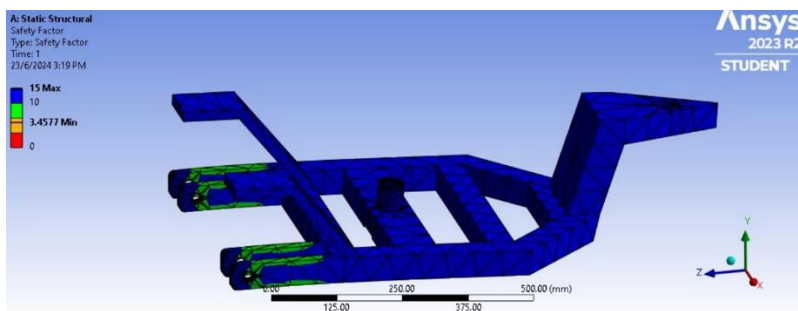
Model (A4) > Static Structural (A5) > Solution (A6) > Total Deformation

Time [s]	Minimum [mm]	Maximum [mm]	Average [mm]
1.	0.	0.40233	7.6615e-002



Model (A4) > Static Structural (A5) > Solution (A6) > Equivalent Stress

Time [s]	Minimum [MPa]	Maximum [MPa]	Average [MPa]
1.	3.662e-005	17.353	2.9286



Model (A4) > Static Structural (A5) > Solution (A6) > Stress Tool > Safety Factor

Time [s]	Minimum	Maximum	Average
1.	3.4577	15.	13.06

Figure 4.8: Ansys results for the frame

Case studies also provided additional insights by analyzing existing lawn mower designs and their performance in real-world applications. Successful models were studied to identify features and design elements that contributed to their popularity such as durability, effective cutting systems, and user-friendly controls. On the other hand, analyzing product failures and recalls highlighted common pitfalls such as weak blade attachments, poor safety features, or inadequate stability on uneven ground. Customer feedback and reviews of existing products were also examined to understand the expectations of end-users and to address issues that had led to dissatisfaction with competing models. This combination of qualitative and quantitative data from surveys, simulations, and case studies formed the foundation for designing a lawn mower that would stand out in terms of performance, durability, and user satisfaction.

4.2 Data Analysis and Interpretation

The data collected through surveys, simulations using ANSYS and SolidWorks, and case studies were analyzed comprehensively to inform and refine the lawn mower design. The survey results indicated several key user priorities. Most respondents preferred lightweight, battery-powered lawn mowers due to their convenience, reduced noise, and environmental benefits. Noise reduction and ergonomic design were highlighted as significant concerns, particularly for residential users. This feedback helped set the baseline requirements for the mower's performance and features.

The CAD models developed in SolidWorks were instrumental in the design process, enabling the visualization of the lawn mower's structure and components. These models facilitated iterative improvements by allowing the team to address design flaws early, such as areas prone to stress concentration or inadequate space for motor components. Once the preliminary design was completed, advanced simulations in ANSYS provided detailed insights into the mower's performance. FEA results showed that the frame and handle could withstand operational stresses but required reinforcement in specific areas such as the joints and mounting points to ensure long-term durability.

Furthermore, case studies validated the simulation results and helped refine the design. Successful lawn mowers in the market emphasized features such as strong materials, user-friendly controls, and effective safety mechanisms. Analyzing product recalls and failures highlighted the importance of secure blade attachments, reliable motor components, and

ergonomic handles to prevent user discomfort or potential safety hazards. For instance, cases of blade detachment in poorly designed mowers reinforced the need for strong fastening mechanisms and material testing. The customer reviews analyzed during the case study phase emphasized common pain points like heavy mowers being difficult to move. By integrating these lessons into the design, the team was able to address known issues proactively and align the product with market expectations.

Data Analysis

Scooter Movement on Road vs Grass

$$\text{Speed} = \text{Wheel} \times \text{Circumference} \times \text{RPM}$$

$$= 2(0.10)255$$

$$= 160.22 \text{ m/min}$$

Converting into km/h:

$$\text{Speed (km/h)} = 160.22 \times 0.060$$

$$= 9.61 \text{ km/h}$$

The speed of a scooter on road versus grass can vary significantly due to differences in surface resistance, friction, and terrain irregularities. Scooter needs more power on an uneven surface like grass to maintain the same speed, while the road has a smooth surface with low rolling resistance. The speed is likely to drop to **5.0-7.0 km/h**, depending on grass type and terrain.

Area Covered Per Hour

$$\text{Cutting speed} = \text{Cutting width} \times \text{Travel speed} \times \text{Efficiency factor}$$

$$= 0.316 \text{ m} (2.67 \text{ m/s}) 0.85(85\%)$$

$$=0.7172 \text{ m}^2/\text{s}$$

Converting into Square Meters per Hour:

$$\text{Cutting speed (m}^2/\text{h)} = \text{Cutting speed (m}^2/\text{s)} \times 3600$$

$$=0.7172 \times 3600$$

$$=2582 \text{ m}^2/\text{h}$$

This lawn mower can mow approximately **2,582 square meters per hour** under the given conditions.

Results Interpretation

For scooter performance, on-road travel is efficient, with constant speeds near the theoretical maximum. Grass significantly impacts performance making the scooter less effective for off-road use without power adjustments or a more robust motor. Hence, for lawn mower performance, the mower demonstrates high productivity, capable of mowing large areas efficiently, making it suitable for residential or light commercial use. Practical limitations like terrain type and operator could slightly affect this performance.

The performance comparisons are shown in the table below:

Category	Scooter (Road)	Scooter (Grass)	Lawn Mower
Speed	9.61 km/h	5.0–7.0 km/h	9.61 km/h (travel speed)
Surface Resistance	Low	High	Moderate (on flat, dry terrain)
Energy Requirement	Optimal	Higher power needed	Consumes moderate energy/power
Efficiency	High	Reduced (~30–50%)	85% (accounting for overlap)
Area Covered Per Hour	Not applicable	Not applicable	2,582 m ² /h (~0.64 acres/hour)
Terrain Suitability	Smooth roads	Uneven grass	Best on flat lawns, struggles on uneven or sloped areas

Table 4.1: performance comparisons

5.0 PROTOTYPE DEVELOPMENT

The Prototype Development phase serves as a critical stage in transforming conceptual designs into tangible, functional models. This process bridges the gap between theoretical planning and practical application by integrating design, fabrication, and testing. For this project, the prototype combines the features of a scooter with a lawn mower to create an innovative and efficient tool for lawn maintenance. This section outlines the steps taken to construct, assemble, and refine the prototype to meet functional and performance requirements. The aim is to validate the design concept while ensuring its usability, safety, and practicality in real-world conditions.

5.1 Materials

1. Blade

The prototype blade is constructed using high-quality steel, selected for its durability and strength. The dimensions of the blade are as follows: a length of 32 cm, a width of 8 cm, and a height of 2 cm. The overall weight of the blade is 0.5 kg, ensuring an optimal balance between robustness and manageable handling characteristics.



Figure 5.1: Blade dimension

2. Frame

The main frame is constructed using galvanized square hollow section steel, chosen for its corrosion resistance and structural integrity. The steel sections have dimensions of 3/4 inch by 3/4 inch, with a thickness of 1 mm, and a length of 2 meters. A total of three sections will be used for the frame assembly.

For the seat support and front support, galvanized steel with larger dimensions and thickness is utilized to provide enhanced strength and stability, as these areas are subjected to greater loads and stresses. The steel sections for the seat and front support measure 3/4 inch by 2 inches with a thickness of 2 mm, and a length of 2 meters. Only one section of this size was purchased for the construction of the seat and front support structures. For the seat post the circular hollow steel with diameter of 26mm and length 300mm was purchased.



Figure 5.2: Galvanized square hollow steel



Figure 5.3: Circular hollow steel

3. Seat

The seat selected for the lawn mower scooter machine is a wide bike seat designed to offer both comfort and durability. It is made from PU leather, a synthetic material that is water-resistant, easy to maintain, and highly durable, making it well-suited for outdoor applications. The seat's interior features thickened high-density foam, which provides excellent shock absorption to minimize the impact of vibrations and uneven terrain during operation. This ensures a smoother and more comfortable experience for the user, particularly when operating the machine for extended periods. With a sleek black color, the seat not only enhances the aesthetic appeal but also effectively conceals dirt and wear, adding to its practicality.

The seat has a weight of 1.15 kg and dimensions as shown in the **figure X..** Its wide design ensures better weight distribution, reducing pressure points and improving user stability. This design choice, along with its cushioning, makes it ideal for long hours of operation. The primary reason for selecting this seat is its ability to absorb shocks, a critical feature for ensuring operator comfort while working on uneven surfaces. Moreover, the PU leather material enhances the seat's durability and resistance to environmental conditions, while its lightweight and versatile design allow for easy installation. These features collectively make this seat a reliable and efficient choice for the lawn mower scooter machine.



Figure 5.4: Seat dimension

4. Throttle

The throttle chosen for the lawn mower scooter machine is a 3-speed electric scooter throttle with forward and reverse functionalities. This throttle is equipped with a handle grip design, ensuring easy and ergonomic operation for the user. The 3-speed control feature allows precise adjustments to the scooter's speed, offering better control over its performance based on the task or terrain. Additionally, the forward and reverse functions enhance maneuverability, making it more convenient to operate the machine in tight spaces or when repositioning is required.

The inclusion of an electric accelerator further ensures smooth and responsive speed transitions, improving the overall user experience. This throttle is compatible with various electric vehicles, such as e-bikes and scooters, showcasing its versatility. Its durable construction and intuitive design make it an ideal choice for enhancing the efficiency and functionality of the lawn mower scooter machine.



Figure 5.5: Throttle with 3 speed buttons

5. Brushless Motor

The motor used in the lawn mower scooter machine is an 8-inch brushless motor designed for electric vehicles, offering durability, efficiency, and smooth operation. With a rated power of 350W and a rated voltage of 24V/36V/48V, this motor provides sufficient torque and speed for the application. Its wheel diameter is 200mm, and the brake system utilizes a drum brake, ensuring reliable and safe stopping power. The motor is capable of speeds ranging from 12 to 28 km/h and can handle a load weight of approximately 80-150 kg, making it suitable for various applications.

This motor is commonly used in electric bicycles, scooters, skateboards, folding bikes, wheelchairs, light electric quad cars, and other DIY projects, showcasing its versatility. The fork open size requirement is 95mm, and it is designed for rear-wheel installation with a drum brake system. It is equipped with a Hall sensor, allowing precise control and efficient power delivery. Additionally, its easy-to-install design enhances convenience during assembly.

Refer to **Figure 5.7** for the motor's dimensions, which are integral to ensuring compatibility with the lawn mower scooter machine. The combination of its power, efficiency, and adaptability makes this brushless motor an excellent choice for the project, providing consistent and reliable performance.



Figure 5.6: Brushless motor

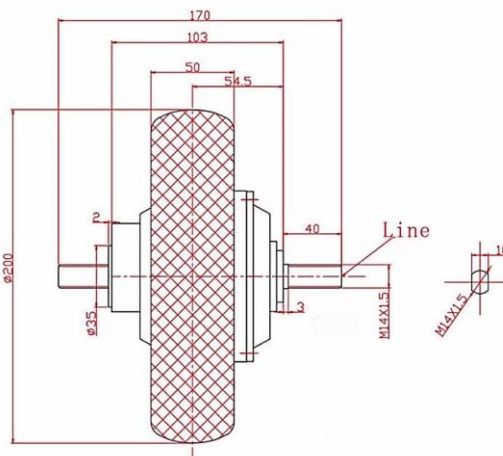


Figure 5.7: Brushless motor dimension

6. Controller

The controller used in the lawn mower scooter machine is the Ausbond brushless motor controller, designed for high-performance electric vehicles such as e-bikes, scooters, and other similar machines. This controller is compatible with 24V, 36V, and 48V systems and is rated for 350W of power, making it capable of handling the motor's performance efficiently under various conditions. Its intelligent control system ensures smooth speed regulation and sensitive handling of braking and directional changes, providing a seamless user experience.

Additionally, the Ausbond controller is equipped with protective features such as undervoltage and overcurrent protection to safeguard the motor and battery from potential electrical faults. It supports E-ABS braking systems, which enhance safety by cutting off power during braking. The controller also incorporates an automatic identification feature for Hall sensors and phase angles, simplifying installation and ensuring compatibility with a variety of motors. With its durable and compact design, the Ausbond controller is built to endure outdoor environments while maintaining reliable and efficient operation, making it an ideal choice for the lawn mower scooter machine.



Figure 5.8: Ausbond Controller

7. Blade Motor

The lawn mower scooter machine is powered by a 24V DC motor with a power output of 300W, making it a reliable and efficient choice for the application. This motor, commonly used in electric scooters, is capable of reaching a maximum speed of

2650RPM under no-load conditions. Its compact design, with a motor diameter of 101mm, allows for easy integration into the scooter's frame, while the provided foot mounting bracket simplifies the installation process.

Although the motor does not include gear reduction, which may result in reduced starting torque, this limitation can be addressed by using a larger chain sprocket to enhance the initial torque performance. The motor comes pre-installed with an 11-tooth sprocket (#25 chain size), making it ready for chain-driven applications. Rated at 14 amps, the motor operates efficiently within the system's 24V power supply. This motor is particularly suited for lightweight electric vehicles, wheelchairs, and even combat robots, making it an excellent choice for the lawn mower scooter machine due to its balance of power, speed, and adaptability.



Figure 5.9: 24V DC motor

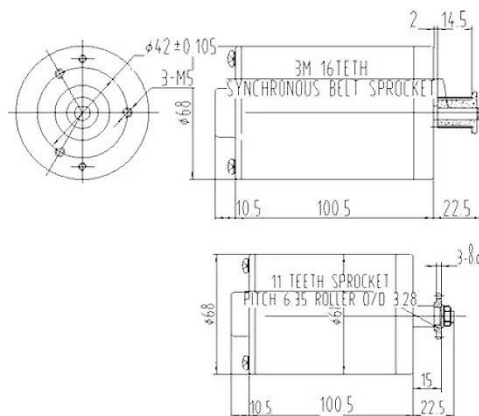


Figure 5.10: Motor dimension

8. Plate

For the plate material, a checkered aluminum steel plate has been chosen due to its ideal combination of strength, weight, and durability, crucial for the application in a lawn mower scooter machine. The plate's dimensions of 600 cm x 500 cm and thickness of 2 mm provide the necessary surface area and rigidity while maintaining a relatively low weight of approximately 3.5 kg. This weight is important in ensuring the scooter remains maneuverable without compromising its structural integrity.

The checkered pattern offers an additional advantage by increasing the surface friction, which can contribute to better traction and stability for the user. In terms of load and stress, aluminum steel is known for its high strength-to-weight ratio, meaning it can withstand significant mechanical stress without adding excessive weight to the overall scooter. This makes it a suitable choice for components that need to endure operational stresses, such as the forces exerted during movement, vibration, and the impact from the mower's operation.

The material's resistance to corrosion also makes it an optimal choice for outdoor equipment like lawn mowers, ensuring longevity even under exposure to weather conditions. Additionally, aluminum steel offers ease of fabrication, which is beneficial for the design and manufacturing processes of the scooter.



Figure 5.11: Aluminium checkered plate

9. Tyre

For the rear tire of the lawn mower scooter, a pneumatic tire has been selected due to its ability to absorb shocks and provide a smooth ride on various terrains. The 8-inch diameter and 5 cm width strike a balance between maneuverability and load-bearing capacity, ensuring optimal performance for the scooter, especially when navigating uneven surfaces commonly found in lawn mowing applications.

Pneumatic tires are known for their flexibility and cushioning properties, which help distribute the load more evenly, reducing the stress on the scooter's frame and providing better traction. This is especially important for a lawn mower scooter, as it needs to perform well across different surfaces, such as grass, dirt, or gravel. The air-filled design also contributes to better grip and stability, ensuring that the tire can handle the dynamic forces experienced during operation without compromising safety.

Additionally, the choice of a pneumatic tire ensures better durability and longer service life. The tire's ability to absorb vibrations minimizes the wear and tear on both the tire and the rest of the scooter's components. This makes the pneumatic tire a reliable and efficient choice for the rear wheel, contributing to the overall performance and longevity of the lawn mower scooter.



Figure 5.12: Pneumatic tyre dimension

10. Battery

For the power supply, the system utilizes four 12V batteries that are combined in a battery box to provide the necessary voltage for the lawn mower scooter's operations. This configuration ensures efficient power distribution between two distinct motors: the blade motor and the brushless motor, each requiring different voltage levels.

The blade motor operates on 24V, which is provided by the combination of two of the 12V batteries wired in series. This arrangement ensures that the blade motor receives the required power to efficiently cut grass or other materials without overloading the system. On the other hand, the brushless motor, responsible for the movement of the scooter, operates on 48V, which is achieved by wiring the remaining two 12V batteries in series. The higher voltage for the brushless motor allows for greater torque and speed, optimizing the scooter's performance on various terrains.

By using this configuration of combined 12V batteries, the system can effectively power both motors independently while maintaining a balanced energy distribution. The separate power supplies for each motor prevent any interference between the two, ensuring that each motor operates at its optimal performance level. This approach also allows for efficient battery usage and extended operational time, contributing to the overall effectiveness and reliability of the lawn mower scooter.

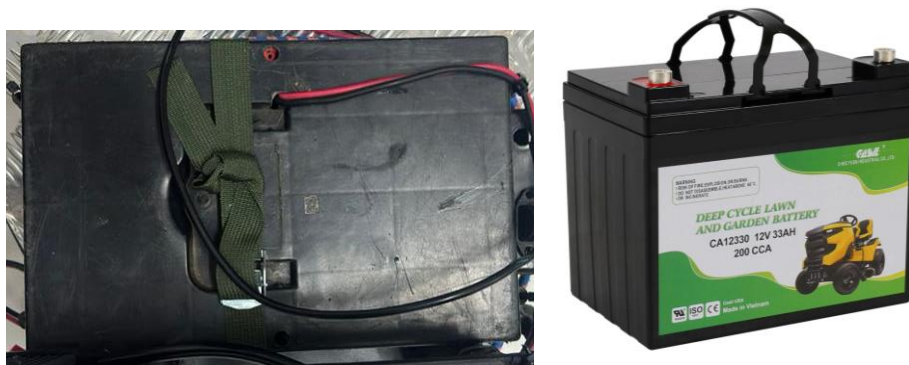


Figure 5.13: Battery box and 12V battery

5.2 Processes

1. Cutting

The frame construction process involves the cutting of the galvanized steel sections to the required lengths. The steel sections for the frame, which were purchased as 2-meter lengths of square hollow section steel (3/4 inch x 3/4 inch, 1 mm thickness), and the galvanized steel sections for the seat and front support (3/4 inch x 2 inches, 2 mm thickness), are cut according to the design specifications. Horizontal band refer **figure 5.14** saw was chosen in the cutting process of this galvanized steel.

Procedure for Cutting the Frame with a Horizontal Band Saw:

1. Preparation

Ensure the horizontal band saw is clean and operational then wear appropriate safety equipment such as gloves and safety shoes. Inspect the steel sections for burrs or defects

2. Marking the Cutting Points

Use a steel ruler to determine the required cutting points, and mark them using a scribe for precise, scratch-like lines on the material.

3. Setting Up the Band Saw

place the steel section on the work table and align the scribed line with the saw blade. Secure the section with the band saw's clamp to prevent movement.

4. Cutting Process

Switch on the machine and carefully lower the blade onto the steel, allowing it to cut smoothly without forcing it. Repeat for additional cuts, ensuring each line is aligned with the blade.

5. Finishing

Inspect the edges of the cuts for smoothness and use a grinder to remove burrs.



Figure 5.14: Horizontal Band Saw Machine

The plate cutting process involves the use of a grinder to achieve the desired shape.

Procedure for Cutting Aluminum Plates with a Grinder:

1. Preparation

Select an angle grinder with a cutting disc suitable for aluminum. Secure the plate on a sturdy workbench using clamps. Make sure the grinder is already plugged in. Wear appropriate safety equipment such as gloves and safety shoes.

2. Marking the Cutting Points

Measure and mark the cutting points on the aluminum plate using a steel ruler. Use a scribe to create clear, accurate scratch marks for cutting.

3. Cutting Process

Secure the aluminum plate on a workbench with clamps to prevent movement. Power on the grinder and guide it steadily along the scribed lines and avoid applying excessive pressure to prevent overheating or deforming the aluminum.

4. Cooling and Inspection

Allow the aluminum plate to cool naturally if it heats up during cutting and
Inspect the cut edges for accuracy and smoothness.

5. Finishing.

Smooth any rough or sharp edges using a grinding disc designed for aluminum.
Lastly clean the plate to remove any dust or debris

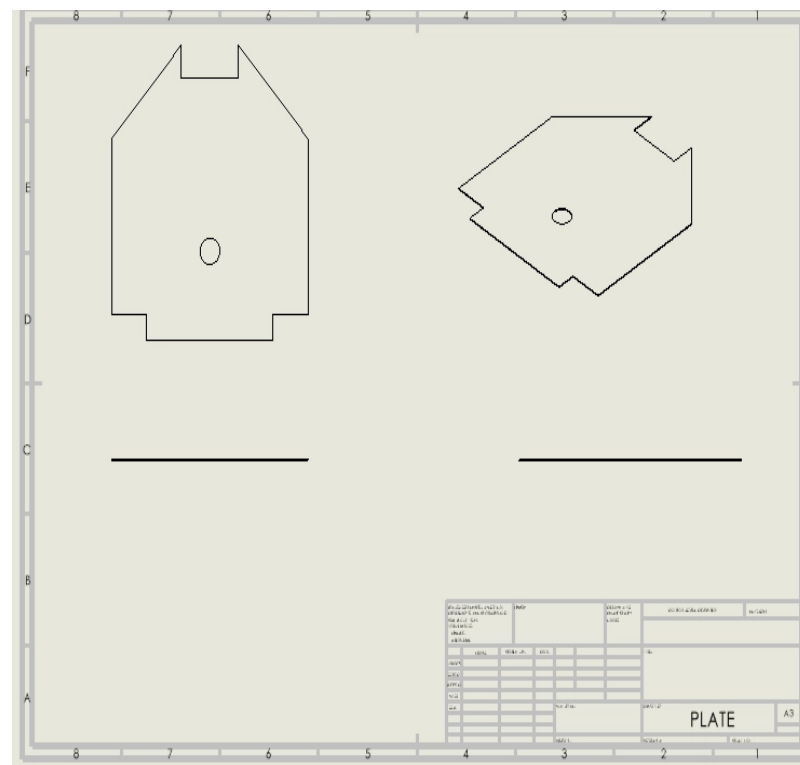


Figure 5.15: Plate Sketch Drawing

2. Welding

The welding process was carried out using the Metal Inert Gas (MIG) technique to join the galvanized steel pieces, which had been precisely cut to size using a horizontal band saw. This approach ensures strong and reliable connections between the components. Tack welding was performed first to ensure the frame's shape aligned with the correct dimension refer **figure 5.16**. Once the frame was properly positioned, full welding was carried out to complete the structure.

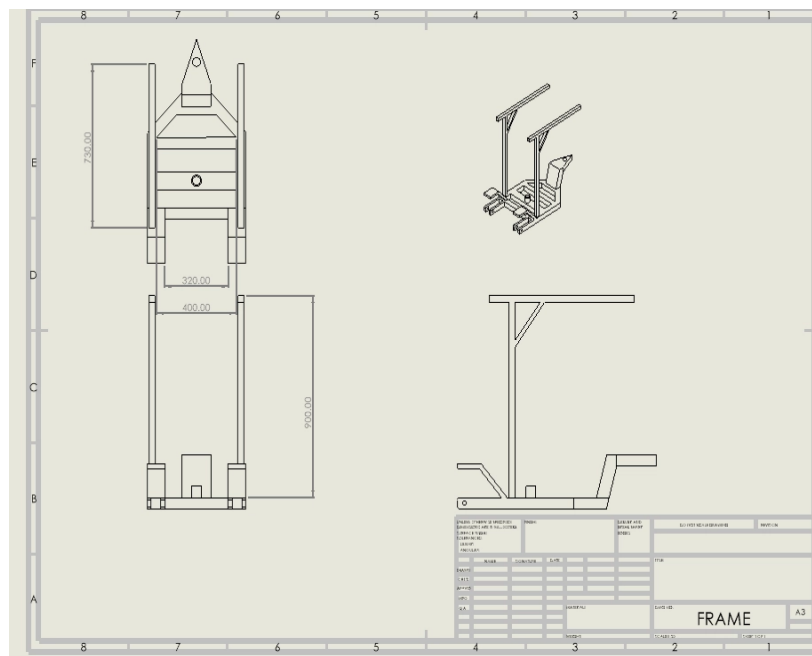


Figure 5.16: shows the desired shape and dimension

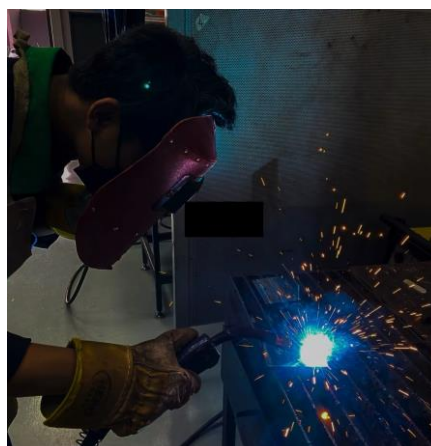


Figure 5.17: MIG welding process

MIG Welding Procedure:

1. Preparation of Materials

First, the metal surfaces must be cleaned thoroughly to remove rust, paint, oil, or debris, often using a wire brush. Make sure the welding base area is also clean to prevent defects such as porosity, weak welds, or poor fusion.

2. Setup of Welding Equipment

The welding equipment is set up by installing the appropriate welding wire, such as ER70S-6 for steel, and selecting a shielding gas mixture. The wire feed speed and voltage are adjusted based on the material's thickness and wire type.

3. Safety Measures

Safety precautions, including the use of personal protective equipment like a welding helmet, gloves, and fire-resistant clothing, are essential, along with proper ventilation to minimize exposure to fumes.

4. Tack Welding

Before proceeding with full welding, tack welding is performed at key points to secure the metal pieces and ensure proper alignment and dimensions.

5. Full Welding

Once the frame is correctly positioned, full welding is carried out by striking the arc, maintaining a consistent travel speed, and keeping a steady arc length. The welding technique, such as pushing or pulling, is selected based on the desired penetration and weld appearance.

6. Inspection

After welding, the joints are inspected for defects like porosity or undercut, and any imperfections are addressed by grinding and re-welding if needed.

7. Post-Weld Cleaning and Final Adjustment

Finally, the weld area is cleaned to remove slag, spatter, or other impurities, and the dimensions of the welded structure are verified to ensure accuracy and alignment.



Figure 5.18: Frame condition after welding process

3. Wiring

The wiring process for this project encompasses the integration of multiple critical components, including the brushless motor, throttle, controller, battery system and blade motor. **Figure 5.19** shows how the flow of the wiring connection of this project.

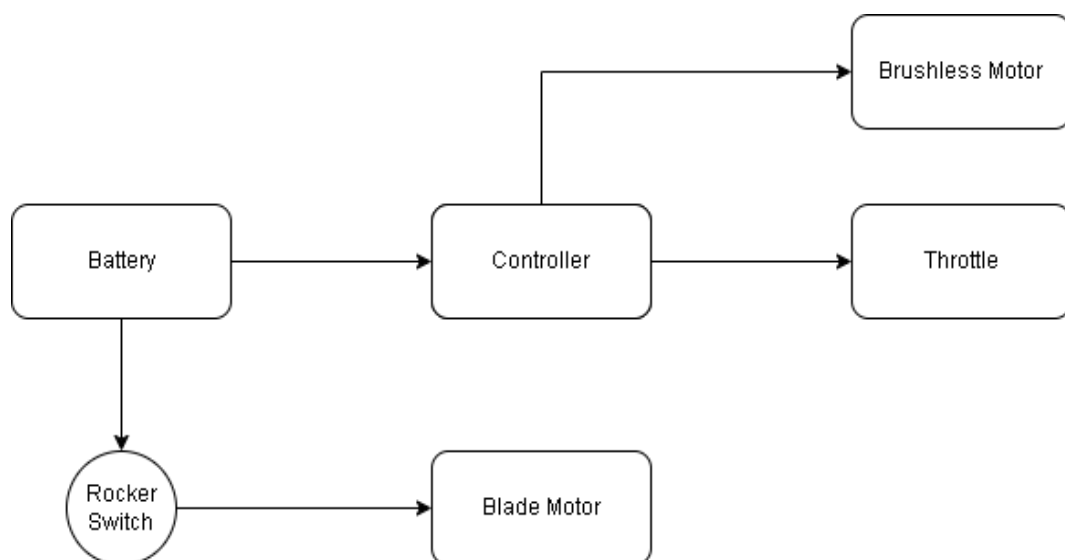


Figure 5.19: Flow of the wiring connection

5.3 Assembly

During the assembly process, the two primary components—the brushless motor (front wheel) and the pneumatic tire (rear wheel)—are installed first on the frame. The brushless motor is secured using M14 washers and nuts, with a size 21 spanner utilized for the installation. Similarly, the pneumatic tire is assembled using M12 nuts, and a 19mm spanner is employed to ensure proper fastening.



Figure 5.21: Assembly process of the brushless motor



Figure 5.22: shows the condition of pneumatic tire and brushless motor after installed

Next, the plate is mounted onto the frame using screws and washers. A total of 15 screws and washers are utilized to securely attach the plate to the frame. Blade motor also got attached to the frame by using 2 screws and nuts (M6).



Figure 5.23: shows condition before install the plate and blade motor

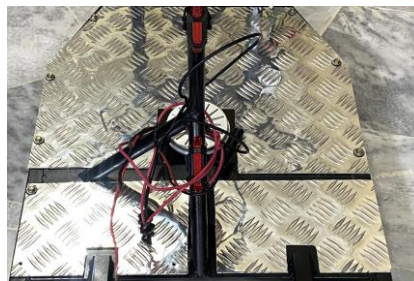


Figure 5.24: condition after plate and blade motor installed

After completing the installation of the blade motor, the blade is mounted using hexagonal bolts (M6) and washers.



Figure 5.25: blade is mounted

The final step of the assembly process involves installing the seat and wiring system components, including the battery, controller, and throttle. To ensure a neat and organized wiring arrangement, cable ties are used to secure the wires, preventing them from being tangled or hanging loosely, which could pose a safety hazard. The battery is fastened with a stretchable rope to keep it securely in place and prevent movement during operation. Additionally, accessories such as the operational indicators light are installed at the rear of the seat.



Figure 5.26: Prototype with fully assemble



Figure 5.27: Operational indicators light

5.4 Cost

COMPONENT	UNIT	PRICE PER UNIT (MYR)
Blade 	1	25.90
Galvanized square hollow steel 	4	5.00
Circular hollow steel 	1	4.50
Seat 	1	28.80

Throttle 	1	23.29
Brushless motor 	1	229.79
Controller 	1	159.00
Blade motor 	1	150.00
Plate 	1	109.64

Pneumatic tyre 	2	30.00
Battery 	4	50

Table 5.1: Cost of component

5.5 Testing Methodology

Movement Testing

Movement testing was conducted on two different surfaces: grass and road. These tests were designed to assess the prototype's ability to maneuver and maintain stability in various environments. The movement on grass proved to be more challenging compared to the road surface. The grass, with its uneven texture and resistance, required more traction and presented additional challenges to the prototype's grip and handling. The road surface, on the other hand, provided a smoother, more consistent terrain, allowing for easier movement and more predictable performance. The theoretical analysis regarding grip and friction were validated during these tests, and the prototype demonstrated satisfactory handling in both conditions, though grass terrain presented a higher level of difficulty.

Theoretical Analysis: Grass vs Road Surface

- Friction on Grass

Grass surfaces generally present higher rolling resistance due to the uneven texture and the natural variability of grass height, density, and moisture content. The grass blades act as an additional layer of resistance against the wheels or blades of the prototype, requiring more force to move the prototype forward. This increased resistance can reduce the effective traction, meaning the prototype may experience slippage or difficulty maintaining consistent speed. The coefficient of friction between rubber (or whatever material is used for wheels) and grass is typically higher than that of a smooth, paved surface, requiring more torque to initiate and maintain movement.

- Friction on Road

In contrast, a road surface provides lower rolling resistance due to its smooth and consistent texture. Asphalt or concrete offers much less friction compared to grass, enabling the prototype to move with less effort. The coefficient of friction between rubber and road is lower, meaning less force is needed to overcome resistance. The prototype is likely to experience less slippage and can maintain a more consistent and predictable speed on a road surface.

Movement Testing Procedure:

1. Preparation:

- Ensure that the prototype is fully assembled and operational, with the blade and movement components in place.
- Select two testing environments: one grassy area with varying grass heights and density, and one smooth, paved road surface.
- Confirm that the weather conditions are consistent across both testing areas. Make sure not too wet or dry to avoid extreme variability in surface conditions.

2. Testing on Grass:

- Begin the movement test on the grass surface by placing the prototype at a starting point.
- Activate the prototype and allow it to move across the grass at a consistent speed.
- Monitor the prototype's stability, grip, and handling. Take note of any slippage, difficulty in movement, or irregularities.
- Record the time taken to cover a specific distance (10 meters) and any performance issues such as reduced speed or uneven motion.
- Repeat the test in different areas of the grass to account for variations in grass density and height.

3. Testing on Road:

- Move the prototype to a paved road surface and ensure the area is clear of obstacles.
- Activate the prototype and allow it to move across the road, repeating the process of tracking speed, stability, and ease of movement.
- Record the time taken to cover the same distance on the road (10 meters).
- Compare the prototype's performance on the road surface to that on grass, specifically focusing on ease of movement, grip, and speed consistency.

4. Data Analysis

- Compare the data collected on grass versus the road, paying close attention to differences in movement speed, stability, and grip.
- Theoretical calculations regarding friction and grip should be validated or adjusted based on real-world performance.

Cutting Testing

The cutting performance of the prototype was tested to ensure that the blade speed is sufficient for effective grass cutting. The tests were conducted on various types of grass, including both dense and thin varieties, to evaluate the blade's effectiveness across different conditions. The prototype successfully cut through the grass, with particular attention given to ensuring that the blade maintained sufficient rotational speed to achieve a clean and efficient cut.

Cutting Testing Procedure:

1. Preparation

- Ensure the blade is sharp and properly mounted on the prototype.
- Select an appropriate grass type for testing. Ideally, use a mixture of dense and thin grass such as fescue, ryegrass to simulate different cutting conditions.
- Set up a designated test area where grass height is consistent across the testing area 5 cm tall grass.

2. Initial Test Run:

- Position the prototype at the starting point of the test area and activate the cutting mechanism.
- Move the prototype across the grass, ensuring that the blade maintains a consistent speed throughout the test.
- Observe whether the blade is able to cut through the grass cleanly and efficiently. Take note of any areas where the grass is not properly cut or where the blade speed seems insufficient.
- Measure the length of grass cut in one pass to assess the efficiency of the prototype.

3. Variation in Grass Types:

- Test the prototype on different types of grass, such as thick, dense grass and lighter, thinner grass.
- Observe and record how the blade performs under these conditions. Note any differences in cutting efficiency, power required, or issues with clogging or blade slowing down.
- Repeat the test in different weather conditions such as dry and slightly wet grass to evaluate the prototype's adaptability to varying environments.

4. Performance Evaluation:

- After completing the tests on different grass types, evaluate whether the prototype maintains sufficient blade speed and cutting efficiency across all test scenarios.
- Check the quality of the cut like whether the grass is neatly trimmed or left uneven.
- Compare the performance to theoretical expectations, ensuring that the blade speed and torque are sufficient to cut through grass without stalling or inefficiency.

6.0 CONCLUSION

6.1 Summary of the report

In conclusion, the riding lawn mower demonstrates promising results in effectively removing grass in the environment. The utilisation of the prototype in simulated conditions showcased its potential to address grass issues, contributing to the environmental conservation efforts. The report highlights key parameters such as debris removal efficiency, operational conditions, and potential applications. Additionally, the report emphasizes the importance of further testing and validation in real-world scenarios to ascertain the scalability and adaptability of the riding lawn mower for broader environmental applications.

6.2 Suggestion and improvement

Improving the braking system on a riding lawn mower can enhance safety and performance. Here are several ways to improve the braking system such as upgrading brake pads and shoes, use hydraulic brakes and regenerative braking. Often, lawn mowers use basic brake pads or shoes that wear down over time. Upgrading to higher-quality materials, such as ceramic or semi-metallic pads, can improve stopping power and reduce wear. If the mower uses mechanical or drum brakes, upgrading to hydraulic brakes could provide smoother and more reliable braking performance. Hydraulic systems generally offer better control and power with less effort. Some advanced mowers now use regenerative braking, where the system recovers energy while slowing down, storing it in a battery for future use. This could be especially useful for electric mowers. If the mower uses drum brakes, consider switching to disc brakes. Disc brakes offer more consistent performance, better heat dissipation, and are less prone to fading. Increasing the size or improving the ergonomics of the brake pedal could make it easier to engage the brakes, especially for people with limited foot strength. We need to ensure that the pedal has a responsive feel and is not too stiff or too loose to improve control. For hydraulic brake systems, regular checking and replacement of the brake fluid can maintain peak performance. Old or contaminated fluid can reduce brake efficiency. If we are using mechanical brakes, upgrading to high-quality brake cables or ensuring the linkage is in good condition will prevent brake fade and ensure immediate engagement when the brake pedal is pressed. Some high-end mowers now include an automatic safety braking system that engages when the operator leaves the seat or the mower slows to a stop. Adding a safety system could

further reduce accidents. While not part of the braking system directly, better traction from high-quality tires can help improve braking performance by reducing slippage.

Using a lamp or an indicator light on a lawn mower to signal that it is operating serves several important safety and operational purposes. The lamp provides a clear indication that the mower is running. This can be helpful in ensuring the operator is aware that the mower is still active, especially in situations where the sound of the engine might not be noticeable (e.g., in noisy environments, on a larger property, or in areas with background noise). The lamp can act as a safety reminder. It ensures that anyone near the mower is aware that it's in operation. This is particularly important when people approach the mower or when kids or pets are around. The mower typically has rotating blades or moving parts that are dangerous when the mower is in operation. A light can remind operators to stay clear of these parts or help them safely approach and use the mower. In electric mowers, the lamp can indicate that the battery is connected and the mower is running. For gas mowers, it could signal that the engine is on. This helps users identify issues like a dead battery or stalled engine. In some cases, if the lamp is part of a system to show when the mower is in use (e.g., a safety interlock system), it ensures that users do not mistakenly attempt to start or engage the mower when it's unsafe, such as when maintenance is being performed. Some regions or manufacturers may include the lamp as part of regulatory requirements for safety features on riding lawn mowers. These standards aim to reduce the risk of injury by ensuring that operators, bystanders, and maintenance personnel are aware of the mower's operational status. In some cases, a lamp can serve as a warning light that signals when there's a malfunction, such as an overheating engine, low oil, or another operational issue that requires attention. This helps prevent further damage to the mower and allows the user to address the problem promptly. So, the lamp essentially provides visual feedback about the mower's operating state, contributing to both user awareness and safety for anyone in proximity to the machine.

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